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Emission probability in vehicle map matching via Hidden Markov models

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Abstract. The focus of this work is factors affecting emission probability in vehicle map matching via hidden Markov models. Commonly used factors are described such as distance, heading, and speed. The importance of rarely considered road width and geometry's representation factors is adjusted. Ways to estimate the emission probability are compared based on different researches. Our own view of estimating the emission probability is presented also.

Keywords. map matching; HMM; emission probability; factors; distance; road width; geometry representation; heading; speed

1. Introduction. Map Matching is a problem of matching sequence of points onto the road graph. Points always have geographical coordinates and timestamp. Sometimes they have azimuth and speed.

Map matching can be applied for pedestrian/in-door navigation, but this work is focused on map matching for vehicles.

Various methods have been proposed to solve map matching problem, and there is still room for improvement. One of the widely used methods is Hidden Markov models due to its efficiency and simplicity.

Hidden Markov models consider both the history of motion and the confidence of mapping the point onto the segment of road graph. The history of motion is presented via so-called *transition probability*. Transition probabilities are not considered in this work. Instead, we focus on confidence of mapping a point onto a road segment which is *emission probability*.

2. Features of emission probability. General approach of estimating the emission probability is independently deriving confidences of mapping by a couple of features such as distance and heading. Usually, the confidence of each feature is estimated in range from zero to one. Then the confidences are multiplied together to get the final confidence of mapping. Below is the overview of such features.

2.1. DISTANCE. The closer a point to a segment, the higher probability of the point to belong the segment.

The simplest solution is to reduce probability linearly by increasing the distance as it is done by Greenfeld (2002) but this approach is rarely used.

Most researches assume the point distances of the segment because of GPS error which can be approximate by a normal distribution. The only parameter of this feature then is the standard deviation σ which varies from 4 (Newson & Krumm (2009)) to 6.86 (Goh et al. (2012)) meters. Our own estimation calculated on Ticon's dataset from more than 10M points of σ is 4.5 meters.

Estimating the confidence of binding by the position, one should also take into account the road width which is described in the section below.

2.2. ROAD WIDTH.

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2.2.1. IMPORTANCE OF THE ROAD WIDTH. Road width is an important factor which is ignored by many of the researches (see table 1). Importance of this factor is depicted by the following observation.

Most researchers consider distance from point to the segment to be normally distributed with standard deviation $\sigma < 7$ meters (see table 1). By the three sigma rule, confidence of point to be related to the segment with distance more than $7 * 3 = 21$ meters is less than 0.3%. Meanwhile, the width of a wide road can excess this value (see Figure 1). Thus, confidence of mapping by position feature could be dramatically underestimated for vehicles on edge lanes if road width is not considered.

Figure 1. Width of a freeway is about 24 meters

By the database of Ticon (2022) we counted Median Absolute Deviation (MAD) of distance between point and the mapped segment depending on lanes number. MAD value is important as it is directly bounded to the standard deviation of the error by factor 1.4826 (Leys et al. (2013)). As it is seen in the Figure 2, MAD (and therefore σ) increases more than twice when the number of lanes growths from one to nine. Thus, ignoring road width when using distance factor could lead to twice underestimation of the emission probability for wide roads such as highways.

The values of MAD in meters are in the Table 1. For each lane, there are more than 30 thousands measurements. N.B. the estimations are taken after applying our own map matching algorithm which can distort the values. Despite, they seem reasonable.

2.2.2. GEOMETRY REPRESENTATION. Natural way of storing road geometry is using a 2-dimensional curve. The nuance here is that the curve could dispose on the corner of the real road shape (see Figure 3) or goes through the center of the shape (Figure 4).

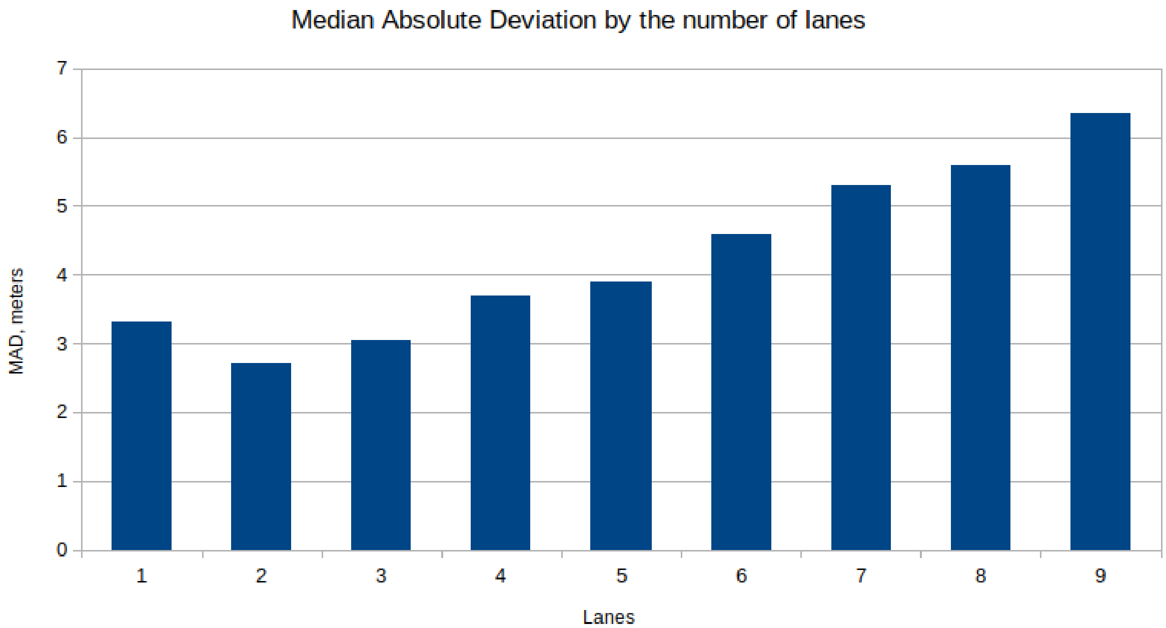

Figure 2. MAD differs a lot depedning on the number of lanes

Lanes	1	2	3	4	5	6	7	8	9
MAD	3.3	2.71	3.04	3.7	3.9	4.58	5.3	5.58	6.34
MAD, Corner line	2.75	2.4	2.93	3.64	3.8	4.5	5.3	5.58	6.34
MAD, Center line	3.63	5.27	4.85	5.4	4.77	6.3	-	-	-

Table 1. Median Absolute Deviation by the number of lanes

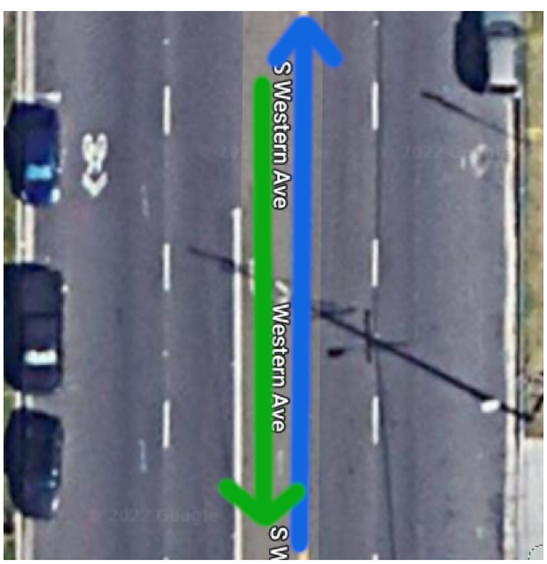

Figure 3. Corner line geometry

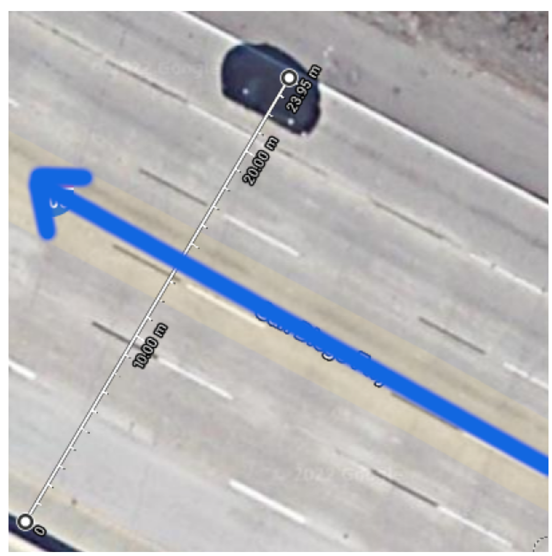

Figure 4. Center line geometry

Ignoring the geometry representation could lead to losing half of the information about the road width which again spoils the confidence of mapping. The MAD's change depending on geometry representation is presented in Figure 5 and Table 1. Based on this data, MAD (and therefore σ) differs from 25% on 5-lanes road to more than 100% for two-lanes road, which reveals the importance of considering the importance of the geometry representation.

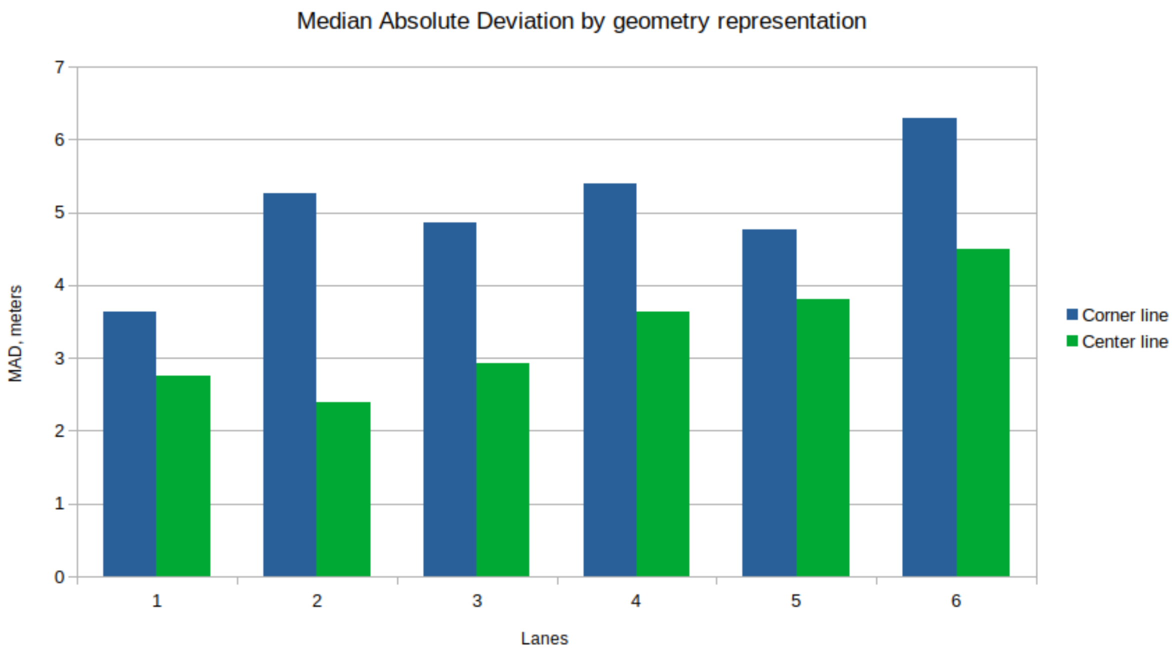

Figure 5. MAD differs a lot depending on the number of lanes

2.2.3. Road width factor. Two estimations are proposed to summarize the distance, road width and geometry representation.

For each number of lanes and geometry presentation own σ of distance error can be used (see Table 1. The disadvantage here is that the estimations are taken based on already implemented mapping without different σ for different lanes taken, so there is a reason to doubt them. σ of wide roads could be underestimated.

Another approach is adjusting distance using formula below:

$$d_{adjusted}(d) = \begin{cases} max(d - width_{road}/2, 0), & \text{if center line} \\ max(d - width_{road}, 0), & \text{if corner line} \end{cases}$$

Then the adjusted distance is passed to the GPS error distribution formula to get the mapping confidence. The deficiency of such formula is assuming that a car is always in the center/corner of a road.

2.3. Radius of interest. It's not possible to consider all segments of the network. Usually, segments from some radius of interest are taken. This value vary in different research from 10 Hansson et al. (2020) or 50 Chen et al. (2019) to 150 Song et al. (2018) and 200 Newson & Krumm (2009) meters.

Increasing the radius slows the algorithm's execution. Decreasing it too much may cause filtering out true segments, so the compromise should be found. In our research radius of 50 meters is taken. It is calculated as following.

Take the median absolute deviation for the roads with the most number of lanes given (nine). Multiply it by 1.4826 to get σ of normal distribution by MAD method, which is described in Leys et al. (2013). Thus, $\sigma = 1.4826 * 6.34 \approx 9.4$ meters. Segments with distance greater 5σ are so unlikely, so there is no need to even consider them. Thus, take $5 * 9.4 = 47 \approx 50$ meters as the radius of interest.

2.4. HEADING. Heading is defined as absolute difference between the azimuth of the point and the azimuth of the road curve. If heading data is not given, some researches suggest deriving it from coordinates between the current and the next points.

There are two main approaches used to estimate confidence of mapping by heading.

The first one is setting probability 1 for heading difference less than threshold $20 - 25°$ (Ren (2012), Hansson et al. (2020)), as this is the angle car turns during a lane change. After the threshold probability linearly goes to zero until the next threshold value about $65 - 90°$.

The second one is using $\cos$ (Greenfeld (2002) Quddus et al. (2003)). It smoothly fades from 1 to 0.94 on headings from 0 to $20°$, taking zero at $90°$ which reasonably approximates probability of the heading deviation.

There is also the third way. One can use standard deviation of angle as it is done for the feature distance Chen et al. (2019). It seems, such approach may lead to mistakes. Most of heading's deviation is caused by natural car's turns more than devices' error. Estimating distribution of heading by single σ may discriminate points taking a turn maneuvers. There will be less such points than others in research's dataset, and they may be considered as an outliers which could lead to less precise results.

2.5. SPEED. The bigger speed limit excess, the lower probability of point to belong to a segment. To penalize the excess of speed limit Goh et al. (2012) propose to use formula $s(v_p, v_l) = \frac{v_l}{max(0, v_p - v_l) + v_l}$, where v_p is the speed of the point and v_l is the speed limit. Thus, twice excess of the speed limit twice decreases the confidence.

That approach could be improved considering that people tend to excess little speed limits more than big ones. In other words, moving 40 km/h with limit 20 km/h is much more likely than moving 200 km/h with limit 100 km/h, so absolute values should be considered. In the approach by Goh et al. (2012) both cases would get factor 0.5 which may be inaccurate. Also, excess of speed limit 4 times is so unlikely, so the segment should not be considered at all meanwhile the formula above would estimate such confidence as 0.25, so such estimation is not strict enough.

In opposite, too strict approach is proposed by Xiong et al. (2021). They set zero probability of mapping if speed excess the limit. Excess of the limit is a common phenomena, so such approach would filter out too many segments. The advantage, though, is that segments with speed limit more than threshold can be skipped at all to improve the performance of the algorithm.

We filter segments with limit less than $v_p/2.41$.

3. Emission probability in different researches. Main characteristics of used featured are presented in Table 2. Most of researches use normal distribution with σ from 4 to 6.86 meters. Road width and speed are rarely used. No one reported about taking into account the geometry representation.

Distance, d	Road width	Heading, h	Speed	Radius, meters	Source
-10 * d	-	$cos^{1.2}(h)$	-	not reported	Greenfeld (2002)
coeff / d	-	$cos(h)$	-	not reported	Quddus et al. (2003)
$\sigma = 4.07$	-	-	-	200	Newson & Krumm (2009)
$\sigma = 4.02$ $(80 - d)/80$	-	$cos(h)$	-	200	Velaga et al. (2009)
$\sigma = 6.5$	-	$1, [0; 25°]$ $line, (25, 65°]$ $0, > 65°$	-	not reported	Ren (2012)
$\sigma = 6.86$	+	-	$\frac{v_l}{max(0, v_p - v_l) + v_l}$	50	Goh et al. (2012)
$\sigma = ?$	road class	empirical	-	not reported	Mohamed et al. (2016)
$\sigma = 5.02$	-	-	-	150	Song et al. (2018)
$\sigma = ?$	-	$\sigma = ?$	-	50	Chen et al. (2019)
$\sigma = 4.07$	+	$1, < 20°$ $else?$	$\frac{v_l}{max(0, v_p - v_l) + v_l}$	10	Hansson et al. (2020)
$\sigma = 2.5$	?	$\cos(h)$	$<$ limit	50	Xiong et al. (2021)
$\sigma = 4.5$	+	$\cos(h)$	$< 2.41*$ limit	50	Ticon (2022)

Table 2. Confidence of mapping by features

4. Conclusions. Factors affecting emission probabilities are distance, heading, and speed. Speed is usually compared with speed limits, heading deviation is considered via cos or custom linear formula. Distance is transformed to confidence of mapping via distribution of GPS error which is approximated as a normal one.

Considering the distance, one should also take into account road width and geometry representation. The quantification of such features' affect has not been researched, but their importance is justified.

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